LAB-03 Solutions

Object Oriented Programming - University of Fallujah, College of Information Technology

Name: _____ Section: _____

Task 01

Part 1 - integer item code

Code

```
#include <iostream>
#include <string>
#include <cstdlib>
#include <ctime>
#include <iomanip>
using namespace std;

const int SIZE = 10;

struct InventoryItem
{
    int code;
    string name;
    double cost;
    double sellingPrice;
};

// takes name and cost from the user and builds one item
InventoryItem insertItem()
{
    InventoryItem item;

    // auto generated code between 3000 and 9999
    item.code = 3000 + rand() % 7000;

    cout << "Enter item name: ";
    getline(cin, item.name);

    cout << "Enter item cost: ";
    cin >> item.cost;
    cin.ignore(1000, '\n');

    // selling price = cost + 40% profit
    item.sellingPrice = item.cost + item.cost * 0.40;

    return item;
}

// displays all items in a table
void displayStock(const InventoryItem* items)
{
    cout << endl;
    cout << left << setw(10) << "Code"
         << setw(20) << "Name"
         << setw(12) << "Cost"
         << setw(12) << "Price" << endl;
    cout << "-----------------------------------------------" << endl;

    for (int i = 0; i < SIZE; i++)
    {
        cout << left << setw(10) << items[i].code
             << setw(20) << items[i].name
             << setw(12) << fixed << setprecision(2) << items[i].cost
             << setw(12) << items[i].sellingPrice << endl;
    }
}

int main()
{
    srand(time(0));
    InventoryItem items[SIZE];
```

```
    for (int i = 0; i < SIZE; i++)
    {
        cout << "--- Item " << i + 1 << " ---" << endl;
        items[i] = insertItem();
    }

    displayStock(items);

    return 0;
}
```

Output

```
--- Item 1 ---
Enter item name: Laptop
Enter item cost: 500
--- Item 2 ---
Enter item name: Mouse
Enter item cost: 10
--- Item 3 ---
Enter item name: Keyboard
Enter item cost: 25
--- Item 4 ---
Enter item name: Monitor
Enter item cost: 150
--- Item 5 ---
Enter item name: Desk
Enter item cost: 200
--- Item 6 ---
Enter item name: Chair
Enter item cost: 90
--- Item 7 ---
Enter item name: Printer
Enter item cost: 120
--- Item 8 ---
Enter item name: Router
Enter item cost: 40
--- Item 9 ---
Enter item name: Cable
Enter item cost: 5
--- Item 10 ---
Enter item name: Speaker
Enter item cost: 35.5

Code      Name                Cost        Price
-----------------------------------------------
4661      Laptop              500.00      700.00
3392      Mouse               10.00       14.00
6471      Keyboard            25.00       35.00
5366      Monitor             150.00      210.00
4166      Desk                200.00      280.00
3541      Chair               90.00       126.00
9829      Printer             120.00      168.00
5881      Router              40.00       56.00
4819      Cable               5.00        7.00
8149      Speaker             35.50       49.70
```

Part 2 - item code as string (e.g. COM_31203)

Code

```
#include <iostream>
#include <string>
#include <iomanip>
using namespace std;

const int SIZE = 10;

// running number used in the item code (first item gets 31203)
int runningNumber = 31203;

struct InventoryItem
{
```

```
    string code;
    string name;
    double cost;
    double sellingPrice;
};

// makes the code: first 3 letters of name in upper case + _ + running number
string generateCode(const string& name)
{
    string prefix = name.substr(0, 3);

    for (int i = 0; i < (int)prefix.length(); i++)
    {
        prefix[i] = toupper(prefix[i]);
    }

    string code = prefix + "_" + to_string(runningNumber);
    runningNumber++;

    return code;
}

// takes name and cost from the user and builds one item
InventoryItem insertItem()
{
    InventoryItem item;

    cout << "Enter item name: ";
    getline(cin, item.name);

    cout << "Enter item cost: ";
    cin >> item.cost;
    cin.ignore(1000, '\n');

    item.code = generateCode(item.name);

    // selling price = cost + 40% profit
    item.sellingPrice = item.cost + item.cost * 0.40;

    return item;
}

// displays all items in a table
void displayStock(const InventoryItem* items)
{
    cout << endl;
    cout << left << setw(14) << "Code"
         << setw(20) << "Name"
         << setw(12) << "Cost"
         << setw(12) << "Price" << endl;
    cout << "-----------------------------------------------------" << endl;

    for (int i = 0; i < SIZE; i++)
    {
        cout << left << setw(14) << items[i].code
             << setw(20) << items[i].name
             << setw(12) << fixed << setprecision(2) << items[i].cost
             << setw(12) << items[i].sellingPrice << endl;
    }
}

int main()
{
    InventoryItem items[SIZE];

    for (int i = 0; i < SIZE; i++)
    {
        cout << "--- Item " << i + 1 << " ---" << endl;
        items[i] = insertItem();
    }

    displayStock(items);
```

```
    return 0;
}
```

Output

```
--- Item 1 ---
Enter item name: computer
Enter item cost: 500
--- Item 2 ---
Enter item name: hard disk
Enter item cost: 60
--- Item 3 ---
Enter item name: Mouse
Enter item cost: 10
--- Item 4 ---
Enter item name: Keyboard
Enter item cost: 25
--- Item 5 ---
Enter item name: Monitor
Enter item cost: 150
--- Item 6 ---
Enter item name: Desk
Enter item cost: 200
--- Item 7 ---
Enter item name: Chair
Enter item cost: 90
--- Item 8 ---
Enter item name: Printer
Enter item cost: 120
--- Item 9 ---
Enter item name: Router
Enter item cost: 40
--- Item 10 ---
Enter item name: Cable
Enter item cost: 5

Code          Name                Cost        Price
-----------------------------------------------------
COM_31203     computer            500.00      700.00
HAR_31204     hard disk           60.00       84.00
MOU_31205     Mouse               10.00       14.00
KEY_31206     Keyboard            25.00       35.00
MON_31207     Monitor             150.00      210.00
DES_31208     Desk                200.00      280.00
CHA_31209     Chair               90.00       126.00
PRI_31210     Printer             120.00      168.00
ROU_31211     Router              40.00       56.00
CAB_31212     Cable               5.00        7.00
```

Task 02

Code

```
#include <iostream>
#include <iomanip>
using namespace std;

struct FeetInches
{
    int feet;
    int inches;
};

struct RoomDimension
{
    FeetInches length;
    FeetInches Width;
};

// converts feet and inches to feet (example: 10 ft 6 in = 10.5 ft)
double toFeet(const FeetInches& f)
{
    return f.feet + f.inches / 12.0;
}

// cost of carpet for one room = length * width * price per square foot
double computeTheCarpetingCost(const RoomDimension& room, const double pricePerSqFt)
{
    double area = toFeet(room.length) * toFeet(room.Width);
    return area * pricePerSqFt;
}

int main()
{
    int rooms;
    double price;

    cout << "Enter the number of rooms in the house: ";
    cin >> rooms;

    if (rooms <= 0)
    {
        cout << "Number of rooms must be greater than zero." << endl;
        return 0;
    }

    cout << "Enter the price of carpet per square foot (Rs.): ";
    cin >> price;

    // dynamic array of rooms
    RoomDimension* house = new RoomDimension[rooms];

    for (int i = 0; i < rooms; i++)
    {
        cout << endl << "Room " << i + 1 << endl;
        cout << "  Length (feet inches): ";
        cin >> house[i].length.feet >> house[i].length.inches;
        cout << "  Width  (feet inches): ";
        cin >> house[i].Width.feet >> house[i].Width.inches;
    }

    int choice;
    do
    {
        cout << endl;
        cout << "========== MENU ==========" << endl;
        cout << "1. Total carpeting cost of the house" << endl;
        cout << "2. Carpeting cost of a single room" << endl;
        cout << "3. Exit" << endl;
        cout << "Enter your choice: ";
        cin >> choice;
        if (choice == 1)
```

```
        {
            double total = 0;
            for (int i = 0; i < rooms; i++)
            {
                total += computeTheCarpetingCost(house[i], price);
            }
            cout << fixed << setprecision(2);
            cout << "Total carpeting cost: Rs. " << total << endl;
        }
        else if (choice == 2)
        {
            int roomNo;
            cout << "Enter room number (1 to " << rooms << "): ";
            cin >> roomNo;

            if (roomNo < 1 || roomNo > rooms)
            {
                cout << "Invalid room number." << endl;
            }
            else
            {
                cout << fixed << setprecision(2);
                cout << "Carpeting cost of room " << roomNo << ": Rs. "
                     << computeTheCarpetingCost(house[roomNo - 1], price) << endl;
            }
        }
        else if (choice != 3)
        {
            cout << "Invalid choice, try again." << endl;
        }

    } while (choice != 3);

    delete[] house;
    house = nullptr;

    return 0;
}
```

Output

```
Enter the number of rooms in the house: 2
Enter the price of carpet per square foot (Rs.): 18

Room 1
  Length (feet inches): 12 0
  Width  (feet inches): 10 0

Room 2
  Length (feet inches): 10 6
  Width  (feet inches): 8 3

========== MENU ==========
1. Total carpeting cost of the house
2. Carpeting cost of a single room
3. Exit
Enter your choice: 1
Total carpeting cost: Rs. 3719.25

========== MENU ==========
1. Total carpeting cost of the house
2. Carpeting cost of a single room
3. Exit
Enter your choice: 2
Enter room number (1 to 2): 2
Carpeting cost of room 2: Rs. 1559.25

========== MENU ==========
1. Total carpeting cost of the house
2. Carpeting cost of a single room
3. Exit
Enter your choice: 2
```

```
Enter room number (1 to 2): 5
Invalid room number.

========== MENU ==========
1. Total carpeting cost of the house
2. Carpeting cost of a single room
3. Exit
Enter your choice: 3
```